Letter to the Editor

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Application of digital pathology in liver transplantation

To the Editor:

Gambella *et al.*[1] proposed a novel approach to assess donor liver steatosis in liver transplantation by integrating deep learning algorithms with Banff consensus recommendations. Applying an automated digital algorithm to a series of donor liver biopsy specimens verifies Banff consensus recommendations' accuracy and practicability. It establishes a standardized dataset for impartially assessing the donor's liver's fat content. Such data would be utilized to optimize the allocation and transplantation effect of the donor's liver. This is an essential application of digital pathology in the field of liver transplantation, and it is a significant contribution. Nevertheless, I would also like to highlight a few deficiencies in the article.

To begin with, the article's sample size is relatively limited, consisting of a mere 292 donor liver biopsy specimens, of which the majority (75%) contain fat content below 30%. This may restrict the generalizability and applicability of this method to donor livers containing varying degrees of lipidosis. Furthermore, additional clinical and pathological characteristics of the donor's liver, including but not limited to age, gender, body mass index, blood glucose, blood lipids, transaminase, and degree of fibrosis, are not mentioned in the article. These factors could impact the quality and efficacy of the donor liver during transplantation.[2] Therefore, I recommend that the authors increase the sample size in future research and incorporate a more significant number of obese liver donors and donor liver characteristics that are more pertinent to the efficacy and robustness of this method.

When investigating the utilization of digital pathology in the liver transplantation domain, we emphasize sample size and representativeness concerns. Prior research has demonstrated the *importance* of sufficient sample size when developing deep learning models that can generalize widely. An investigation examining the application of deep learning to identify microsatellite instability in colorectal tumors, for instance, analyzed 8,836 tumor samples.[3] A deep learning model was trained using data from 1,225 patients *with* gastric cancer in another study.[4] The model was subsequently validated using data from 753 patients *with* gastric cancer in two cohorts. These examples demonstrate the importance of selecting a sample set that is both sufficiently large and diverse to guarantee the validity and accuracy of AI models when assessing liver pathology.

Furthermore, the article's primary conclusion relies on the substantial correspondence between deep learning algorithms and Banff consensus recommendations rather than being compared to conventional evaluation standards for liver transplantation. Regarding the donor risk index[5] or the donor liver survival index,[6] is there a correlation between the lipid content of the donor liver and either statistic? Is there a correlation between the lipid content of the donor liver and various long-term outcomes following liver transplantation, including but not limited to chronic transplant liver disease, early transplant dysfunction, primary non-function, patient survival rate, and acute rejection? Clinically, these are more significant concerns and serve as crucial criteria for assessing the method's clinical significance and applicability. Consequently, I propose that the authors conduct further research to juxtapose this method with these conventional liver transplantation evaluation standards to illustrate its merits and drawbacks.

The article's most notable contribution is its utilization of deep learning algorithms for segmenting and evaluating donor liver fat. However, the article must comprehensively explain the algorithm's principles and implementation and conduct a comparative analysis with other established digital pathology methods. For instance, are there alternative image analysis techniques that yield comparable or superior outcomes, such as conventional image processing methods that rely on threshold, morphology, texture, color, and so forth, or sophisticated image segmentation techniques that utilize alternative deep learning architectures like Mask R-CNN,[7] DeepLab,[8] and so on? Can the algorithm be applied to various image conditions, including staining, magnification, and resolution? Does the algorithm demand substantial computational resources and quantity of annotation data? Does the algorithm support visualization and interpretability? These are critical areas of concern and challenges within the domain of digital pathology, and these issues are also crucial components in assessing the algorithm's scientific and technical merits. Hence, I propose that the authors expound upon the algorithm's underlying principles and implementation in subsequent investigations and conduct a comparative analysis with alternative digital pathology techniques.

Moreover, in contemporary research, obtaining replication and transparency in AI modeling is an urgent challenge that must be resolved immediately. To accomplish this, we suggest implementing a set of standardized criteria for input data (*e.g.*, how images are obtained, processed, and stored), processing criteria (*e.g.*, data segmentation, performance evaluation metrics, model optimization strategies), and output result criteria (*e.g.*, how AI model prediction outcomes are interpreted and represented). The process of staining standardization by employing GAN in the study by Gambella *et al.*[1] ensured color and luminance uniformity across slides, a crucial step in enhancing model performance and dependability. Implementing these standardization measures will not only improve the study's transparency, *but* it will also facilitate the replication and validation of findings by other researchers, thereby propelling the field of digital pathology in liver transplantation evaluation forward.

In conclusion, the article by Gambella *et al.* is a significant endeavor, as it introduces a novel approach to assessing donor liver steatosis during liver transplantation and a fresh concept for the practical implementation of digital pathology in this domain. Conversely, the article does possess certain deficiencies that necessitate refinement and perfection in subsequent investigations.

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Authors' contributions

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The data underlying this article will be shared by the corresponding author on reasonable request. No data were generated during the writing of this study.

Supplementary data

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